



Touché recognises touch on surfaces and complex gestures of the human body

one end and a computer at the other, which analyses the changing signals to identify the particularly gesture in use.

The researchers say Touché could be used to create smart doorknobs that unlock when grasped in a certain way. It could also allow tables and chairs to sense the position of people using them.

Gesture-controlled ambient interface for devices

A new system named Ambient Interface will allow users to interact between devices with the use of gesture control and movement. Developed by Japanese technology giant NEC, it is capable of projecting the image of an input device on any kind of surface to carry out the desired commands. For this model to work, users need a projector and a moveable camera that identifies 3D shapes and movement of hands or fingers.

This virtual input device can help users control all their gadgets with gestures.

According to a Crazy Engineers report, the focus is on achieving a wide range of high-precision control over devices that perceive user inputs to operate. Of late, several cloud-based services have enabled users to explore new avenues of information terminals. With this new gadget, NEC will allow users to control these terminals based on their gestures and movements without interacting with a physical remote control device.

27W LED to replace 100W bulb

A 27-watt LED bulb that can output 1600 lumens or more has been announced. This bulb can replace the standard 100-watt incandescent bulb. It has been made possible due to the development of a method for moving air to cool LEDs using an oscillating membrane, called synthetic jet (an alternative to a fan), which fits within the envelope of the A-19 bulb shape.

The bulb has been developed by GE

in collaboration with Nuventix. GE testing confirms 100-watt incandescent replacement metrics of 1600+ lumens, uniform omnidirectional light distribution, 3000K colour temperature, 25,000-hour life rating (which comes down to 22.8 years at three hours of use per day), dim ability, no mercury, instant full brightness and 60+ lumens per watt—all fitting neatly in the standard A-19 bulb shape.

The product will hit the US and Canada markets by the first half of next year.

Electronic nose sensor to diagnose illness

Soon a smartphone attachment, when used in conjunction with an app, could detect any illness you may have. It will work on what the developers call 'sensory vapour technology.'

It is hoped that the research will result in an affordable, easy-to-carry detection system—ideally an advanced smartphone app. A doctor could carry around a smartphone and have patients breathe into its little attachment and be able to tell whether they have a communicable disease like tuberculosis.

The detection system works by pushing a stream of air through a liquid, such as butanol, which results in bubbles that come up as a saturated vapour. This is then diluted and fed to a sensor chamber where the vapour is analysed and turned into raw data. From this data the scientists are able to work out the chemicals contained in the initial stream of 'air.'

Electronic nose for harmful chemicals

Scientists in the UK have developed an electronic nose sensor that could detect harmful chemicals such as volatile organic compounds. The sensor finds use in safety and health areas.

A new type of electrically conductive material called quantum tunneling generates an electric current when bent or touched. For the electronic nose, this material is made in powder form and placed on films, paper or clothing. Its sensor has low power requirements and advanced materials allow the sensor to detect chemicals



Terminal control based on gestures and movements without interacting with a physical remote control device